Review

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \underline{\qquad\qquad\qquad\qquad} \quad \text{(Left Hand Sum)}$$

$$= \lim_{n\to\infty} \underline{\qquad\qquad\qquad\qquad} \quad \text{(Right Hand Sum)}$$

where $\Delta x =$ _____.
Suppose $f(x)$ is decreasing and continuous. Then _____ is an overestimate and _____ is an underestimate.

The Accuracy of the LHS and RHS

We stated above that the LHS and the RHS both approach the same quantity (the area between the curve and the x-axis, with area below the axis considered negative) as $n \to \infty$. Let's show that they indeed do that, and along the way, get an idea of how accurate our two estimates are.

Let RHS_n and LHS_n denote the right-hand and left-hand Riemann sums respectively, both with n subintervals.

We're trying to show that as n approaches ∞, the RHS_n and LHS_n approach the same number. If that's the case, what should $\lim_{n\to\infty} |RHS_n - LHS_n|$ approach?

Let's compute: $|RHS_n - LHS_n|$
$= |$_____

$-$_____$|$

$= |f(a + ___\Delta x)\Delta x - f(___)\Delta x|$
$= |f(___)\Delta x - f(___)\Delta x|$
$= |f(___) - f(___)|\Delta x$
Now, what happens to Δx as n approaches ∞? What does that tell you the above difference approaches as $n \to \infty$?

1. Suppose we were trying to estimate $\int_1^4 x^2\,dx$.

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(a) To start with, let's use $n = 3$ subintervals.

i. Compute by hand the LHS, RHS, and the difference between them.

ii. Compute $|f(b) - f(a)|\Delta x$ in this case. Check that it matches your previous answer.

(b) If we use $n = 6$ subintervals, what will be the difference between the LHS and the RHS?

(c) What about $n = 60$ subintervals?

(d) How big would you have to make n in order to get a difference of less than 0.01?

2. For each of the following integrals, compute how many subintervals we would need to □Regrade get the difference between the LHS and RHS to be less than 0.01 in absolute value:

(a) $\int_0^{\frac{\pi}{2}} \sin x\,dx$

(b) $\int_0^2 xe^{-x}\,dx$

A Few More Integrals

3. (a) Evaluate $\int_{-3}^{3} 2x\ dx$ by drawing a picture.

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(b) Evaluate $\int_{0}^{3} 3x - 7\ dx$ by drawing a picture.

(c) Evaluate $\int_{-1}^{1} \sqrt{1-x^2}\ dx$ by drawing a picture. (Hint: If $y = \sqrt{1-x^2}$, what is $x^2 + y^2$? What shape is that?)